

**NuraVolt**

TECHNICAL WHITEPAPER

# **BESS Reliability 2025: Preventing Thermal Runaway & Protecting Warranties**

How early warning indicators can prevent \$2M+ damage  
and warranty invalidation in battery storage systems

**Based on research from:**

NREL • Sandia National Labs • EPRI • DNV GL • UL 9540A Testing Data

# The Warranty Challenge

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**Key Finding:** 70% of BESS warranty claims are partially or fully denied due to operator failure to meet monitoring and maintenance requirements specified in manufacturer warranties.

Battery Energy Storage Systems (BESS) come with comprehensive warranties covering capacity degradation and component failures — typically 10-15 years or 10,000-15,000 equivalent full cycles. However, these warranties contain critical monitoring requirements that most operators unknowingly violate.

## Common Warranty Invalidation Triggers

- **Insufficient Monitoring Frequency:** Warranties require continuous monitoring of cell voltage, temperature, and SOC, but many operators only check daily summaries
- **Delayed Response to Warnings:** Manufacturer alert thresholds often require response within 24-48 hours; operators miss these windows
- **Temperature Excursions:** Cell temperatures exceeding manufacturer limits (typically 45-50°C) even briefly can void warranty
- **SOC Imbalance:** Persistent cell-level SOC drift >5% indicates BMS failure but often goes undetected
- **Cycle Counting Errors:** Inadequate cycle counting methodology leads to warranty disputes over remaining capacity guarantees

## Financial Impact of Warranty Denial (10 MWh system):

- Thermal runaway incident: \$2M-\$5M in equipment damage + fire suppression + business interruption
- Premature capacity degradation: \$500K-\$1.2M if warranty claim denied (8-12% below guarantee after 5 years)

- Battery replacement cost: \$300-\$500/kWh = \$3M-\$5M for full 10 MWh system
- Lost revenue during downtime: \$50K-\$100K per month for utility-scale arbitrage applications

## Thermal Runaway: The Catastrophic Risk

Thermal runaway is a self-sustaining exothermic reaction where a cell's internal temperature rises uncontrollably, leading to fire, explosion, and cascading failure across adjacent cells and modules.

### The Progression of Thermal Runaway

| Stage                        | Temperature | Observable Indicators                                    | Action Window  |
|------------------------------|-------------|----------------------------------------------------------|----------------|
| 1. Early Warning             | 40-60°C     | Cell temperature >5°C above neighbors, voltage imbalance | Hours to days  |
| 2. SEI Breakdown             | 60-90°C     | Accelerated self-discharge, voltage drop, gas generation | 30-120 minutes |
| 3. Separator Melting         | 90-135°C    | Internal short circuit, rapid voltage collapse, smoke    | 5-15 minutes   |
| 4. Electrolyte Decomposition | 135-180°C   | Venting, toxic gas release, visible flames               | <5 minutes     |
| 5. Thermal Propagation       | >180°C      | Adjacent cell failures, rack-level cascading             | Seconds        |

**Critical Insight:** Stage 1 is the only intervention window where monitoring can prevent catastrophic damage. By Stage 3, automatic fire suppression is the only defense — and it often fails to prevent total system loss.

# Early Warning Indicators: The Preventive Approach

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Preventing thermal runaway requires detecting precursor conditions hours or days before the temperature excursion begins. The following indicators provide actionable early warning when monitored continuously.

## 1. Cell-Level Voltage Imbalance

**What to Monitor:** Voltage difference between highest and lowest cells within a module.

**Normal Range:** <50 mV during operation, <100 mV at rest

**Warning Threshold:** >100 mV during operation or increasing trend >10 mV/week

**Why It Matters:** Voltage imbalance indicates cell-level degradation, internal short circuit development, or BMS control failure. Cells with lower voltage are forced to over-discharge during operation, accelerating degradation and thermal stress.

## 2. Temperature Gradient Across Cells

**What to Monitor:** Temperature difference between hottest cell and module average.

**Normal Range:** <3°C under normal charge/discharge conditions

**Warning Threshold:** >5°C sustained for >15 minutes, or any cell >45°C

**Why It Matters:** Hotspots indicate localized internal resistance increase from dendrite formation, connection failures, or cooling system inadequacy. Thermal runaway almost always begins at the hottest cell.

## 3. State of Health (SOH) Degradation Rate

**What to Monitor:** Capacity fade rate measured monthly via full charge/discharge cycles.

**Normal Range:** 1-3% per year for LFP, 2-5% per year for NMC chemistry

**Warning Threshold:** >2x expected degradation rate or sudden 5%+ drop in 3 months

**Why It Matters:** Accelerated degradation suggests abuse conditions (temperature, voltage, charge rate) or manufacturing defects that increase thermal runaway risk. Warranty claims also depend on accurate SOH tracking.

#### 4. Charge/Discharge Efficiency

**What to Monitor:** Round-trip energy efficiency (discharge energy / charge energy).

**Normal Range:** 88-95% depending on chemistry and C-rate

**Warning Threshold:** <85% or declining trend >3% over 6 months

**Why It Matters:** Efficiency loss indicates internal resistance increase from cell degradation. Lost energy becomes heat, raising thermal stress and accelerating degradation in a vicious cycle.

#### 5. SOC Drift Between Cells

**What to Monitor:** State of Charge difference between cells after equalization cycles.

**Normal Range:** <2% after full equalization charge

**Warning Threshold:** >5% persistent drift despite equalization attempts

**Why It Matters:** SOC drift that can't be equalized indicates cell-level capacity loss or BMS sensor failure. Operating with imbalanced SOC forces weaker cells into over-charge/discharge stress conditions.

##### **Example: 5 MWh C&I BESS in Dubai**

##### **Early Indicators Detected (Month 18):**

- Module 7 showed 8°C hotspot during 1C discharge (normal: <3°C)
- Cell voltage spread 150 mV in same module (normal: <50 mV)
- SOH degraded 12% vs 4% expected for 18 months operation

**Action Taken:** Module isolated and replaced under warranty before thermal event.

**Outcome:** \$80K warranty replacement vs \$2M+ potential thermal runaway damage and business interruption. Warranty approved because continuous monitoring data proved operator diligence.

# Monitoring Best Practices

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## Essential Monitoring Requirements

| Parameter                    | Resolution | Frequency   | Retention |
|------------------------------|------------|-------------|-----------|
| Cell Voltage (all cells)     | 1 mV       | 1-5 seconds | 10 years  |
| Cell Temperature (all cells) | 0.1°C      | 1-5 seconds | 10 years  |
| Module Current               | 0.1 A      | 1 second    | 10 years  |
| SOC (module level)           | 0.1%       | 1 second    | 10 years  |
| SOH (module level)           | 0.1%       | Monthly     | Lifetime  |
| Ambient Temperature          | 0.5°C      | 1 minute    | 5 years   |

## Alerting Strategy

### Tier 1: Immediate Action (24-hour response required)

- Any cell >48°C or >5°C above module average
- Cell voltage imbalance >150 mV during operation
- SOC drift >8% despite equalization
- Charge/discharge efficiency <80%

### Tier 2: Investigation Required (7-day response)

- Temperature gradient >3°C sustained for >1 hour
- Voltage imbalance >100 mV or increasing >5 mV/week
- SOH degradation >1.5x expected rate for 3 consecutive months
- Efficiency declining >2% over 6 months

### **Tier 3: Monitoring Trend (monthly review)**

- Temperature trends approaching warning thresholds
- Voltage imbalance increasing but  $<100$  mV
- SOH degradation within 1.5x expected but accelerating

### **Data Analysis Techniques**

**1. Baseline Establishment:** Use first 3-6 months of operation to establish normal performance baselines for temperature, voltage spread, and efficiency under various operating conditions.

**2. Peer Comparison:** Compare module performance against other modules in same system to identify outliers. Modules with consistent 10%+ worse performance need investigation.

**3. Environmental Normalization:** Adjust temperature and efficiency metrics for ambient temperature, charge rate, and SOC level to isolate degradation from operating condition effects.

**4. Trend Analysis:** Track monthly moving averages for key metrics to detect gradual degradation that day-to-day noise obscures.

**5. Predictive Modeling:** Use historical data to forecast SOH at warranty milestones and identify modules at risk of falling below capacity guarantees.

# Operational Best Practices to Minimize Risk

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## Temperature Management

- **HVAC Optimization:** Maintain cell temperatures 20-30°C during operation; <35°C at rest. Every 10°C increase doubles degradation rate.
- **Derating Strategy:** Implement power derating at 40°C (reduce to 80% power) and 45°C (reduce to 50% power) to prevent thermal stress.
- **Seasonal Adjustments:** Reduce charge rates by 20-30% during summer months in GCC climates where ambient >40°C is common.
- **Night Charging:** For non-time-critical applications, prefer charging during cooler night hours to reduce thermal stress.

## Charge Management

- **SOC Operating Window:** Limit daily operation to 20-80% SOC; avoid holding at >90% or <10% for extended periods.
- **C-Rate Limits:** Stay below 0.5C charge rate and 1C discharge rate for LFP; 0.3C/0.5C for NMC to minimize degradation.
- **Equalization Frequency:** Perform full equalization charge (to 100% SOC, hold until voltage spread <20 mV) monthly for LFP, weekly for NMC.
- **Calendar Aging:** If system idle >7 days, maintain SOC at 50% and perform shallow charge cycle weekly to prevent lithium plating.

## Preventive Maintenance

- **Quarterly Inspection:** Visual inspection for signs of overheating (discoloration, thermal damage), connection integrity, HVAC operation.
- **Semi-Annual Calibration:** Verify BMS voltage and temperature sensor accuracy against reference instruments ( $\pm 10$  mV,  $\pm 1^\circ\text{C}$ ).
- **Annual Capacity Test:** Full charge/discharge cycle at 0.2C to measure actual capacity vs nameplate for SOH validation.

- **Connection Torque:** Verify busbar and terminal connection torque annually (thermal cycling causes gradual loosening).

#### **Example: 20 MWh Utility BESS in Oman**

**Operational Challenge:** Summer ambient temperatures  $>45^{\circ}\text{C}$  caused cell temperatures to approach  $50^{\circ}\text{C}$  during 1C discharge.

**Solution Implemented:**

- Reduced discharge rate to 0.7C during peak ambient (11am-4pm)
- Shifted arbitrage charging to 10pm-6am (cooler ambient)
- Upgraded HVAC capacity by 30% for better cooling

**Results (12 months):**

- Average cell temperature reduced from  $47^{\circ}\text{C}$  to  $38^{\circ}\text{C}$  during operation
- SOH degradation: 2.1% vs 4.5% projected under previous operating conditions
- Zero thermal events; all modules within warranty capacity guarantees at Year 3 inspection

# ROI from Proactive Monitoring

| Benefit                                            | Annual Value (10 MWh system)      |
|----------------------------------------------------|-----------------------------------|
| Avoided thermal runaway incident (10% probability) | \$200K-\$500K expected value      |
| Warranty claim success (capacity degradation)      | \$50K-\$120K over system lifetime |
| Extended system life (2-3 years)                   | \$150K-\$300K NPV                 |
| Reduced insurance premiums (proof of monitoring)   | \$10K-\$25K annually              |
| Optimized maintenance (predictive vs reactive)     | \$15K-\$30K annually              |
| Total Potential Benefit                            | \$425K-\$975K over 10 years       |

**Cost of Implementation:** \$30K-\$80K for advanced monitoring system + \$10K-\$20K annual analytics service = <10% of potential benefit even in low-probability scenarios.

## Implementation Checklist

1.  **Verify Current Monitoring:** Confirm you have cell-level voltage and temperature data at <5 second intervals with 10-year retention

2.  **Review Warranty Requirements:** Document specific monitoring and response requirements from manufacturer warranty (often buried in technical appendices)
3.  **Establish Baselines:** Use first 3-6 months of operation to define normal performance ranges for your specific system and climate
4.  **Deploy Automated Alerting:** Implement tiered alerting system (immediate, investigation, trending) with documented response protocols
5.  **Schedule Preventive Maintenance:** Quarterly inspections, semi-annual sensor calibration, annual capacity testing
6.  **Document Everything:** Maintain comprehensive records of all monitoring data, alerts, investigations, and maintenance actions for warranty protection

### Protect Your BESS Investment

Get a free 15-minute BESS health assessment from our UAE-based energy storage specialists.

We'll review your current monitoring setup and identify warranty compliance gaps.

[Schedule Free Assessment →](#)

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